

P@SHA ICT AWARDS 2026 · STUDENT INNOVATION CATEGORY

ConceptIntel

A Knowledge Graph–Based Concept Intelligence Platform for Adaptive Higher Education

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Rethinking Digital Learning

Digital learning platforms are widely used across universities, but most focus on managing courses, distributing content, and recording grades — administrative tools rather than intelligent learning systems. They do not analyze how knowledge is structured or how students actually develop conceptual understanding.

ConceptIntel is an AI-powered concept intelligence platform for both teachers and students. It organizes course content as interconnected concepts in a knowledge graph, provides adaptive learning paths using Agentic AI, and generates personalized learning materials using Generative AI — while keeping teachers in the loop to review, edit, and approve every AI-generated output for full transparency.

3

Pilot Subjects

5

User Roles

12

Planned Modules

1

Unified Platform

THE PROBLEM

The Challenge in Higher Education

- ⚠️ Current LMS platforms manage courses and grades — but never measure conceptual understanding.
- ⚠️ Students progress through courses without mastering prerequisite concepts, causing gaps that compound over semesters.
- ⚠️ Teachers see overall scores, not which specific concepts a student is struggling with.
- ⚠️ Generic AI chat tools explain topics but stay disconnected from the actual curriculum.
- ⚠️ Knowledge built by one instructor rarely survives their departure — no institutional reuse.

Student → LMS → Grades



No concept-level visibility for teacher or student.

Student → AI Chatbot → Generic Answer



Not tied to the course, curriculum, or prior mastery.

Existing Solutions vs. ConceptIntel

	Course Management	AI Assistance	Knowledge Graph	Adaptive Learning	Teacher Review	Concept Analytics
Google Classroom	✓	—	—	—	✓	—
Moodle	✓	—	—	—	✓	—
Coursera	✓	—	—	~	—	—
Khan Academy	—	—	—	~	—	—
ChatGPT	—	✓	—	—	—	—
ConceptIntel	✓	✓	✓	✓	✓	✓

Introducing ConceptIntel



AI-Powered Academic Intelligence

One platform serving both students and teachers, built around concepts — not just courses.



Knowledge-Graph-Driven Learning

Course material becomes an interconnected map of concepts and prerequisites.



Retrieval-Augmented Generation

Concept extraction and content search grounded directly in each course's uploaded material.



Adaptive Learning Engine

Personalized paths that reinforce exactly the concepts a student is weak on.



Teacher-in-the-Loop

Every AI-generated concept and knowledge graph is reviewed and approved before students see it.



Curriculum Intelligence

Knowledge graphs are shared and reused across instructors teaching the same course.

GOALS

Objectives

1 Develop a web-based AI platform supporting both students and teachers.

3 Track concept-level mastery for every student.

5 Give teachers actionable insight into learning gaps and bottlenecks.

7 Reinforce weak concepts through personalized, gamified exercises.

2 Extract key concepts from course materials and organize them into a knowledge graph.

4 Generate adaptive learning materials — flashcards, study guides, quizzes, and assignments — with explainable feedback.

6 Enable teacher supervision and approval of all AI-generated content and recommendations.

8 Support continuous content improvement through feedback-driven regeneration.

Prototype Scope

ConceptIntel is being developed as a prototype web-based learning system spanning three subjects, chosen specifically to demonstrate prerequisite-based concept relationships across a curriculum:



Applied Physics

Foundational prerequisite subject



Digital Logic Design

Builds on Applied Physics concepts



Calculus & Analytical Geometry

Independent third pilot subject

- Teachers upload slides, PDFs, textbooks, CLOs, PLOs, reference books, and HEC outlines.
- If full content isn't available, initial concept structure can be generated from outlines alone.
- Knowledge graphs are shared across teachers teaching the same course, and reused by future instructors.
- The project prioritizes prototype validation over full-scale deployment.

End-to-End Workflow



Core Features



Knowledge Graph Builder

LIVE



Content Upload + RAG

LIVE



AI Concept Extraction

LIVE



5-Tier Role-Based Access

LIVE



Coordinator Approval

LIVE



Class Stream

LIVE



Assignments

LIVE



Notification Center

LIVE



Adaptive Learning Engine

UPCOMING



Analytics Dashboard

UPCOMING



Gamification

UPCOMING



Assignment Evaluation

UPCOMING

System Modules (Part 1 of 2)

7.1 User Authentication

Sign-up, login, and role-based access for 5 roles; secure sessions.

LIVE

7.2 Course Management

Create/edit/delete courses with metadata, scheduling, and status.

LIVE

7.3 Enrollment

Enrollment codes, join links, and prerequisite checks.

LIVE

7.4 Content Upload

Upload materials; AI + RAG pipeline extracts concepts and objectives.

LIVE

7.5 Knowledge Graph

Build, visualize, and edit concept dependencies; shared across instructors.

LIVE

7.6 Content Generation

Generate MCQs, flashcards, study guides, and quizzes from concept nodes.

UPCOMING

System Modules (Part 2 of 2)

7.7 Teacher Supervision

Review, modify, and approve AI-generated content and knowledge graphs.

LIVE

7.8 Assignment Evaluation

Evaluate PDF/Word assignments at concept level with explainable feedback.

IN PROGRESS

7.9 Adaptive Engine

Observe-Analyze-Plan-Act loop personalizing each student's learning path.

UPCOMING

7.10 Analytics Dashboard

Concept-mastery heatmaps and bottleneck insights for teachers.

UPCOMING

7.11 Gamification

Mini-games tied to concept nodes, adapting to student mastery.

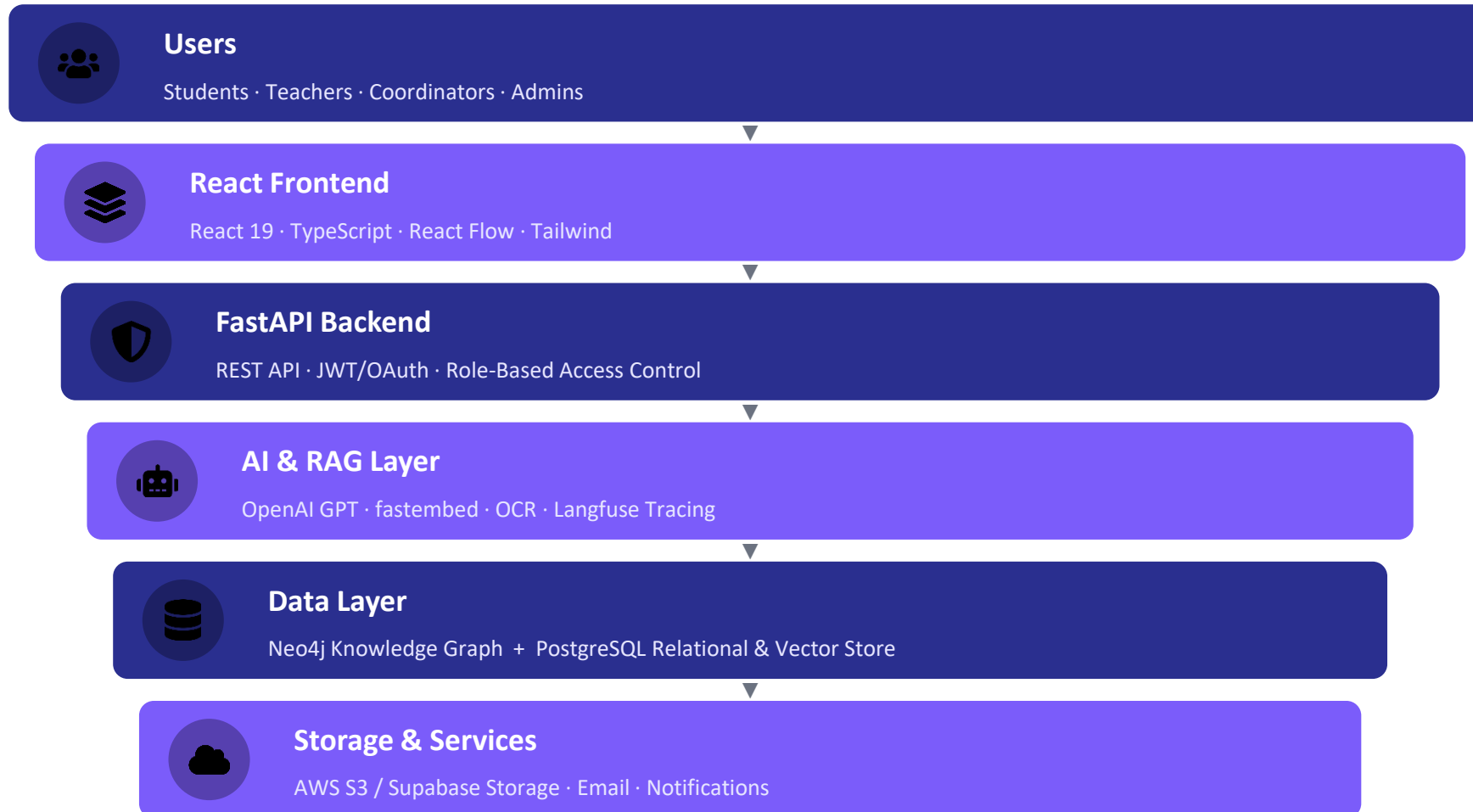
UPCOMING

7.12 Notification Module

Announcements and alerts for new content, quizzes, and feedback.

LIVE

System Architecture



Technology Stack

Frontend

- React 19
- TypeScript
- TanStack Query
- React Flow
- Tailwind CSS
- Vite

Backend

- FastAPI
- SQLAlchemy 2.0
- PostgreSQL
- Uvicorn
- Pydantic

AI / ML

- OpenAI GPT
- fastembed
- Tesseract OCR
- PyMuPDF
- Langfuse Tracing

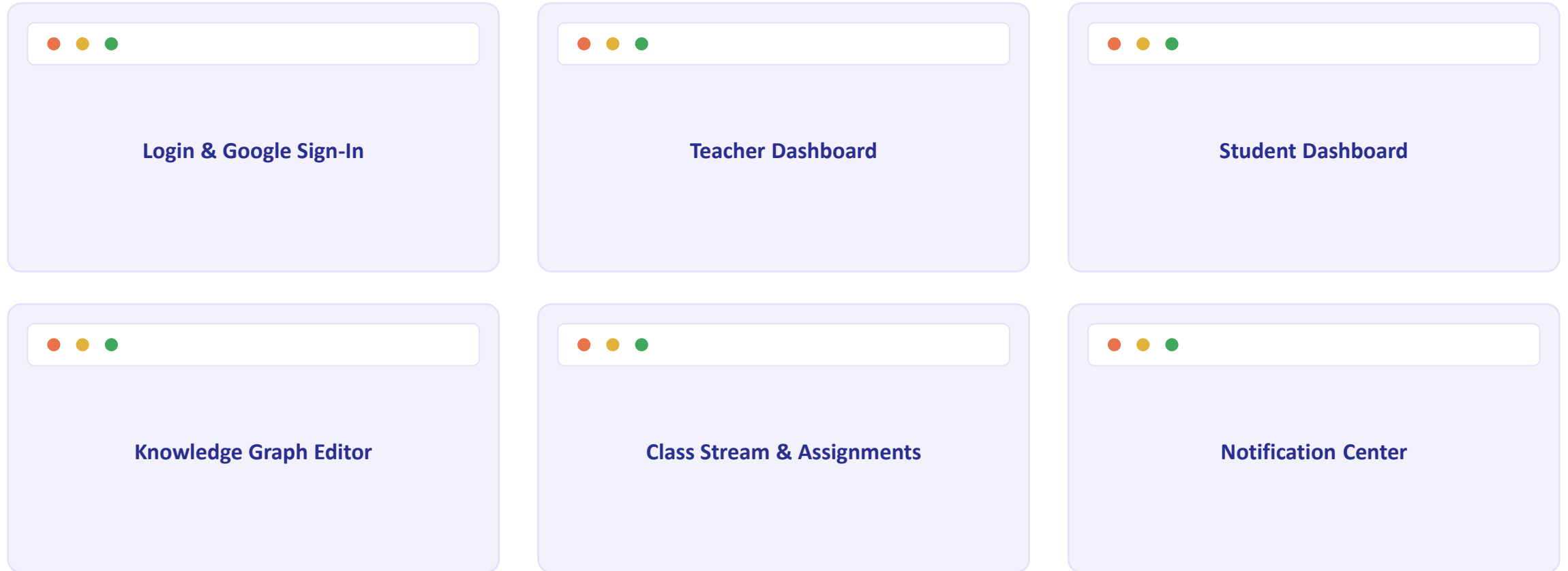
Graph & Data

- Neo4j
- Vector Similarity Search
- SQLAlchemy ORM

Infrastructure

- AWS S3
- Supabase Storage
- Supabase Auth
- Google OAuth
- JWT / bcrypt

Product Demonstration



Prototype currently under active development — screens shown are representative UI flows.

Why ConceptIntel Is Different



AI + Knowledge Graph Fusion

Concept extraction is grounded in a real graph database, not a flat AI response.



Curriculum-Aware RAG

Every answer and generated concept is retrieved from the course's own material.



Teacher-Approved AI

A dedicated Course Coordinator role reviews and approves every knowledge graph.



Shared Institutional Knowledge

Graphs are reused across instructors teaching the same course, term after term.



Multi-Tier Academic Governance

Five distinct roles — Admin, Program & Course Coordinators, Teacher, Student — mirror real university structure.



Concept-Level Analytics

Insight into exactly which prerequisite concepts a cohort is weak on, not just grades.

CURRENT STATUS

Development Progress

Completed

- Multi-Role Auth (5 Roles) + Google Sign-In
- Course & Catalog Management
- Enrollment Module
- Content Upload + RAG Pipeline
- Knowledge Graph Engine (Neo4j + AI)
- Coordinator Approval Workflow
- Class Stream & Announcements
- Assignments (Submission Flow)
- Notification Center
- PostgreSQL Migration

In Progress

- Assignment Evaluation Module (grading & feedback)
- Adaptive Learning Engine (OAPA loop)

Upcoming

- Concept-Level Analytics Dashboard
- Gamification Module
- Deployment & Pilot Testing

System Limitations & Constraints



AI Extraction Isn't Perfect

Automated concept extraction may not always identify concepts perfectly and can require teacher adjustment.



Prototype-Focused

The system targets prototype functionality across 3 pilot subjects rather than full-scale deployment.



Compute at Scale

Large-scale real-time adaptation for many concurrent students may require more computational resources.



Depends on Material Quality

Concept extraction quality depends directly on the quality of uploaded educational materials.

Data Gathering Approach



Educational Materials

Lecture slides, textbooks, and course outlines are processed with AI techniques to extract conceptual information.



User Interaction Data

Quiz responses and learning progress are collected to analyze conceptual understanding over time.



Supervisor Guidance

Continuous guidance from academic supervisors shapes system design decisions.



Existing System Analysis

Analysis of existing digital learning platforms informs feature and workflow design.

Project Stakeholders & Roles



Project Sponsor: Air University, Islamabad



Students

System end-users following adaptive learning paths



Teachers

Course creators and knowledge-graph supervisors



Program & Course Coordinators

Curriculum oversight and content approval



Project Supervisor

Dr. Sumaira Farid — technical and academic guidance



FYP Evaluation Committee

Final assessment of project deliverables

Module-Based Work Division

Team Member	Assigned Modules	Key Responsibilities
Mamoona Latif Reg# 231662	User Management, Content Upload, Content Generation, Notification	Implements authentication, manages content uploads, develops AI content generation, and notifications.
Laiba Reg# 231627	Course Management, Knowledge Graph, Analytics Dashboard, Gamification	Develops course management system, builds knowledge graph, implements analytics, and gamified learning.
Muhammad Kashif Reg# 231568	Enrollment, Teacher Supervision, Assignment Evaluation, Adaptive Engine	Implements enrollment system, AI supervision workflow, assignment evaluation, and adaptive learning engine.

Development Gantt Chart



VISION

Impact, Scalability & Roadmap



Students

Personalized, concept-level learning paths



Teachers

AI-assisted teaching with full oversight



Universities

Reusable curriculum intelligence



Administrators

Data-driven academic decisions



References

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THE TEAM

Meet the Team



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Reimagining Higher Education through Knowledge Graphs and Artificial Intelligence